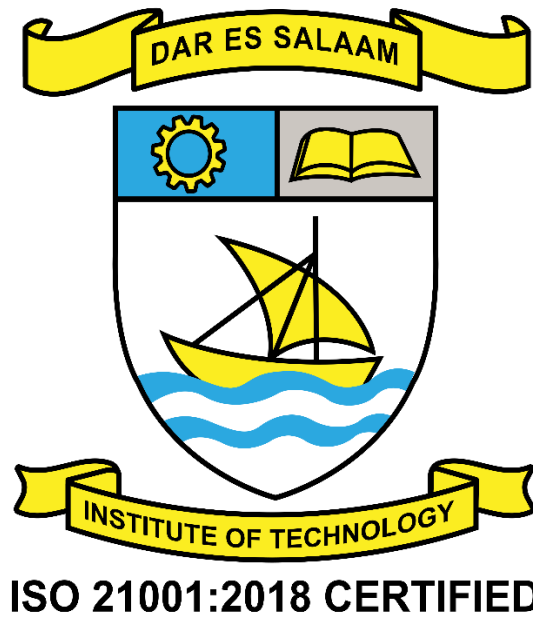


DAR ES SALAAM INSTITUTE OF TECHNOLOGY



DEPARTMENT OF COMPUTER STUDIES

COT 05214 Lab Manual

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LABORATORY SAFETY GUIDELINES

A. General Laboratory Safety Rules

Personal Safety

- Be familiar with the electrical and fire hazards associated with your workplace.
- Be as careful for the safety of others as for yourself. Think before you act.
- Be tidy and systematic.
- Avoid bulky, loose or trailing clothes.
- Avoid long loose hair.
- No one is allowed to enter in the lab area bare foot due to increased risk of electric shock.
- Remove metal bracelets, rings or watchstraps when working in the laboratories.
- Avoid working with wet hands and clothing.

Food, Beverages and Smoking

- Due to the increased risk of electric shock
- No drinking of beverages, consumption or storage of any kind of food is allowed in the laboratory.
- Smoking is prohibited in all laboratories in all timings.

Power Supply Related Safety

- Voltages above 50-VAC or 120-VDC are always dangerous. Extra precautions should be considered as voltage levels are increased.
- Never make any changes to circuits or mechanical layout without first isolating the circuit by switching off and removing connections to power supplies.

Laboratory Equipment

- Lab equipment may not be removed from the Computer lab areas without the permission of the laboratory Supervisors.
- Laboratory bench equipment (except for some lab bench computers) must be turned off before closing down the lab area for the day.
- Never open (remove cover) of any equipment in the laboratories.
- x Never "jump," disable, bypass or otherwise disengage any safety device or feature of any equipment in the laboratories.
- Laboratories shall be locked when unoccupied.

Equipment Safety

- Before equipment is energized ensure, circuit connections and layout have been checked by laboratory supervisors and all colleagues in your group has given their consent.
- Experiments left unattended should be isolated from the power supplies. If for a special reason, it must be left on, a barrier and a warning notice are required.
- Equipment found to be faulty in any way should be reported to the lab supervisor and taken out of service until inspected and declared safe.

Equipment Accessories

- Use extension cords only when necessary and only on a temporary basis. x Request new outlets if your work requires equipment in an area without an outlet.
- Discard damaged cords, cords that become hot, or cords with exposed wiring.

B. Electrical and Fire Emergency Responses**Police, Fire or Medical Emergency**

- Use the telephone located in the laboratory area to notify police, fire, and ambulance for emergency help.
- Everyone present in the laboratory area shall be familiar with the locations and operation of safety and emergency equipment, including but not limited to, fire extinguishers, first aid kits, emergency power off system, fire alarm pull stations, and emergency telephones.

Electric Shock and Electrical Fire

- When someone suffers serious electrical shock, he may be knocked unconscious.
- If the victim is still in contact with electrical current, immediately turn off the electrical power source.
- If you cannot disconnect the power source, depress the Emergency Power Off switch.
- Do not touch a victim that is still in contact with a live power source; you could be electrocuted! Have someone call for emergency medical assistance immediately.
- Administer first-aid, as appropriate.
- If an electrical fire occurs, try to disconnect the electrical power source, if possible.
- If the fire is small and you are not in immediate danger; and if you have been properly trained in fighting fires, use the correct type of fire extinguisher to extinguish the fire.
- When in doubt, push in the Emergency Power Off button.
- NEVER use water to extinguish an electrical fire.

Building Evacuation in Emergency

- Everyone present in the laboratory should be familiar to emergency exits & way out plans.
- Use the nearest exit doorway from lab area closest to the stairwell to exit the building.
- Follow the Emergency Exit Signs posted in the hallways.
- Do not use elevators.

C. LAB GENERAL INFORMATION

C.1 Introduction to the Labs

Each lab will be conducted as per instructions from the subject facilitator. The lab's objective is to introduce students to the practical world of microprocessor technology which includes its organization and architecture. This will enable the students to become familiar with the architecture of microprocessor instruction set. It will provide a hands-on practical experience on assembly language programming and interfacing of various peripheral devices with microprocessor.

Students will also be familiarized with the procedure of designing, simulating and implementing assembly programs. The labs do not give you step-by-step instructions on how to do them. They require you to think carefully about how you are going to do them.

You will do the labs with the 8085 or 8086 simulation software or microprocessor training kit (*as well as interfacing modules for training purposes*) this simulation software and of training kits.

C.2 Lab rules

- Be regular and disciplined: Students are expected to attend all lab sessions regularly.
- Come to the lab sessions on time. Action may be taken to late comers.
- The due date for any lab report is one week after the lab is done. Submit your lab reports on time. Any late report, for no valid reason, will not be accepted.
- Lab is yours, so help to keep it clean. Bringing foods, water or newspaper are forbidden.
- Bring the Lab manual on every time you come to the laboratory.
- Do not make walking around, chatting, or noise during lab period.
- Do not change your lab group or bench without taking a permission from the facilitator.
- Do not do anything in a hurry. Ask the instructor if you don't understand any point or step.
- After finishing the experiment, disconnect the connection wires that you used during the experiment, and put them in the wire kit, take out all IC chips from the training kit and put them back in the IC Cabinet. Be sure to return all materials you used, to clean your table, and to inform the Lab instructor when you depart.

C.3 Grading:

The lab group work assignment grading will be presented to you in the first lecture during the discussion of study guide.

C.4 Requirements for Lab Reports:

Lab reports should be submitted as per deadline date set by the module master which will include all of the following sections:

- Cover Page: The cover page of your lab report should include the lab name(s), your registration numbers and your name.
- The procedures that you follow to do the lab.
- Discussion and Conclusions: In this section, you present the results you got during the lab, your observations about these results, your comments, and your conclusions.

D Materials

D.1 Learning materials

- Marker pens, chalks and projector\smart board for multimedia presentation.
- Interactive simulations software (licensed GNUsim8085, EMU8086).
- Computer lab with desktop computers or laptop

D.2 8085 Microprocessor training Kits.

- Microprocessor training kits MPS 85-3 training kit with its interfacing modules for training modules purpose such as keyboard, elevator, display, analog to digital converter(ADC) with Digital to analog converter(DAC), Dual DAC, logic controller, Traffic light, 8253 demo, RTC, Tone generator, stepper motor, Numeric printer, 8bits-16bits channel ADC, 12bits-16bits channel ADC, opto isolated input, opto isolated output, LCD interface, Relay output, DC motor, RTD interface. Study card for 8255,8279,8259 and 8251/8253, cycle step interface and audio tape interface.
- Power supply: ESA PS M2(SMPS): [+5V@3.0A](#), 12V@250mA, +30V@100mA
- Power Adapter: ADP DC5: [+5V@3.0A](#) SMPS)
- EPROM programmer interface to program 2716 through 27512.

D.3 8086 Microprocessor training Kits.

- 80186 Microprocessor Trainer IT-4319 kit with experiments includes: Controlling the Processor I/O Ports, Address Decoding of an I/O Device, Study of UART 0/1 Device, Controlling Keypad and Joystick, Direction Control of DC Motor, Control of 8-Bit A/D Converter, Control of 8-Bit D/A Converter. The kit should have interfacing addon modules cards such as: Programmable Peripheral Interface (82C55), 16-8 Dot Matrix, 128×64 Graphical LCD, 8-CH Relay Output, DC Motor, R/C Servo, Stepper Motor with Direction Control, Strain Gauge & Temperature Module, Audio Unit, Ultrasonic and Traffic Light
- Power supply: ESA PS M2(SMPS): +5V@3.0A, 12V@250mA, +30V@100mA
- Power Adapter: ADP DC5: +5V@3.0A SMPS)

Lab 1 Arithmetic and Logical Operations**Lab 1.1 A program to add, subtract, multiply and divide two 8 bit numbers****Aim:**

Performing assembly language Programming of addition, Subtraction, multiplication, division using 8085/8086 Microprocessors.

Training Materials Required:

- Microprocessor simulation software
- Computers / laptops
- Trainer kit.

Objective: after completion of this lab a student will be able to

1. Understand the assembly instructions.
2. Perform arithmetic and logic operations.
3. Perform memory operations.
4. Draw a timing diagrams.

Lab 1.1 preamble: The two numbers to be Added, subtracted, Multiplied or divided are stored in memory locations 0801 and 0802. First number is brought to accumulator and the second number in the memory is added, subtracted, multiplied or divided from it. The result of accumulator should be stored in memory location 0803 and carry in memory location 0804. Your program should start at memory address 0000.

Required to do.

1. Draw a flow chart for each operation that shows the steps required to accomplish each operation.
2. Write your program by using the format given below

Address	Label	Mnemonics	Hex code	Remarks

3. Enter the program in step 2 to the simulation software or trainer kit and execute the program you entered.
4. What are the values stored in accumulator, memory location 0803 and 0804?
5. Draw a timing diagram for addition instruction.
6. Write your comments on the final value of flag register and instruction counter
7. Conclude this practical lab 1.1.

Lab 1.2 Assembly language program to add, two 16-bit numbers.**Training Materials Required:**

- Microprocessor simulation software
- Computers / laptops
- Trainer kit.

Objective: after completion of this lab a student will be able to

1. Understand the assembly instructions.
2. Perform arithmetic and logic operations.
3. Perform memory operations.

Lab 1.2 preamble: The two numbers to be Added, locations 0805 and 0806. First number is brought to accumulator and the second number in the memory is added, be stored in memory location 0807 and carry in memory location 0808. Your program should start at memory address 0000.

Required to do.

1. Draw a flow chart for each operation that shows the steps required to accomplish the task.
2. Write your program by using the format given below

Address	Label	Mnemonics	Hex code	Remarks

3. Enter the program in 2 above to the simulation software or trainer kit and execute the program you entered.
4. What are the values stored in accumulator, memory location 0805, 0806, 0807 and 0808?
5. Write your comments on the final value of flag register and instruction counter
6. Conclude this practical lab 1.2.

LAB 2 Memory Operations and Indexing

LAB 2.1 Sorting of 6 numbers in ascending order

Training Materials Required:

- Microprocessor simulation software
- Computers / laptops
- Microprocessor Trainer kit.

Objective: after completion of this lab a student will be able to

1. Use Increment and decrement operations in register and register pairs.
2. Comparison operation.
3. Rotation operation.
4. Looping operation

Lab2.1 preamble:

In this program we will arrange the numbers in bubble sorting technique which uses different pass. on each pass numbers are compared and each largest number is stored at the end of the list. The numbers are stored 2040H to 2046H. The you should store array size at 2040H.

Assume the inputs of the program are-

Address	Data
2040	06
2041	05
2042	08
2043	07
2044	03
2045	06
2046	01

Required to do.

1. Draw a flowchart for each operation that shows the steps required to accomplish the task.
2. Write your program by using the format given below

Address	Label	Mnemonics	Hex code	Remarks

3. Enter the program in 2 above to the simulation software or trainer kit and execute the program you entered.
4. What are the values stored in accumulator, from memory location 2041 to 2046
5. Write your comments on the final value of HL register.
6. Conclude this practical lab 2.2.

LAB 2.2 Searching the Smallest Number in a Given Array

Training Materials Required:

- Microprocessor simulation software
- Computers / laptops
- Microprocessor Trainer kit.

Objective: after completion of this lab a student will be able to

1. Understand microprocessor instruction sets.
2. Perform search operations in array.
3. Rotation operation.
4. Looping operation

Lab2.2 preamble:

To Search the smallest number in a given array the program should compare each value in the array with all values stored in the array.

Assume that the values are stored in memory location 2001-2006. The memory address 2000 hold the size of the array as depicted in the table below

Address	Data
2000	06
2001	55
2002	22
2003	44
2004	11
2005	33
8006	66

Required to do.

1. Write the steps by step algorithm to accomplish searching smallest number.
2. Write your program by using the format given below

Address	Label	Mnemonics	Hex code	Remarks

3. Enter the program in 2 above to the simulation software or trainer kit and execute the program you entered.
4. What are the values stored in accumulator, from memory location 2001 to 2006?
5. Write your comments on the final value of HL register.
6. Conclude this practical lab 2.2.

LAB 3 Interfacing

LAB 3.1 Stepper Motor

Training Materials Required:

- Microprocessor trainer kit.
- Stepper motor interfacing module.
- Study card for 8255,
- Regulated power supply

Objective: after completion of this lab a student will be able to

1. Understand I/O interfacing.
2. Interface stepper motor.
3. Program stepper motor to rotate forward and reverse direction.

Lab3.1 preamble:

A practical 8085 stepper motor control uses an 8255 PPI to send a 4-step sequence to a driver circuit, rotating the motor CW or CCW. A delay routine between steps manages speed.

Required to do.

1. Write the steps to setup hardware and connections for this practical.
2. Write the step by step algorithm for forward and reverse rotation.
3. Write your program by using the format given below

Address	Label	Mnemonics	Hex code	Remarks

4. Enter the program in 2 above to the simulation software or trainer kit and execute the program you entered.
5. Why you should send the control word 80H to the address 03H or C003H?
6. Conclude this practical lab 2.2.

LAB 3.2 Traffic Light Control**Training Materials Required:**

- Microprocessor trainer kit.
- Traffic light interfacing module.
- Study card for 8255.
- Regulated power supply
- Relay board.

Objective: after completion of this lab a student will be able to

1. Understand I/O interfacing.
2. Interface traffic lights
3. Program traffic lights.
4. Use subroutine in the program

Lab3.2 preamble:

The traffic light arrangement is as shown in Fig.1. The traffic should be controlled in the following manner.

1. Allow traffic from W to E and E to W transition for 20 seconds.
2. Give transition period of 5 seconds (Yellow bulbs ON)
3. Allow traffic from N to S and S to N for 20 seconds
4. Give transition period of 5 seconds (Yellow bulbs ON)
5. Repeat the process.

NB. You are required to use a delay subroutine for delaying of yellow, green and red light

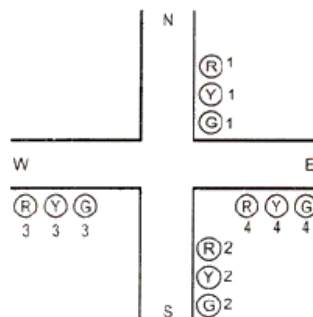


Fig 1.

Required to do.

1. Write the steps to setup hardware and connections for this practical.
2. Draw a flowchart for traffic light operation.
3. Write your program by using the format given below

Address	Label	Mnemonics	Hex code	Remarks

4. Enter the program in 2 above to the simulation software or trainer kit and execute the program you entered.

5. Observe the sequence of traffic light and modify the delay time to 2minutes for traffic transition and 50seconds for yellow light delay
6. Conclude this practical lab 3.2.

